Author:
Francisco Zampella
francisco.zampella@huawei.com

Live positioning App demonstrator

Final assignment, sponsored by Huawei

Embedded Mobile and Wireless Systems (EWireless) 2026 by Prof Tughrul Arslan

The University of Edinburgh

1 Contents

2 Description

During your course you have learned about Android APIs for different functionalities, including creating UI elements, displaying location on a world map, and acquiring sensor data. This final assignment will take you a step further from toy examples to actually using the data and exploring its possibilities in indoor positioning context.

The assignment will be to extend the PositionMe app to include a positioning demonstrator that will fuse the movement model of the user with the GNSS position, the WiFi position and the map of the building. The position should be displayed on the maps of the target buildings and should provide a smooth experience that describe the actual position of the user corrected according to the observed data.

The output we are expecting from the students are:

1. PowerPoint presentation (20 min) of their work process, app functionality, solutions, and results
2. Source code matching the presentation

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The PositionMe app has four core components/functionalities, and in this assignment the student should integrate 1 component, and add 2 new components to improve the experience. The components are:

- WiFi Sampling/Positioning: Creates fingerprints from the observed WiFi and uses the openpositioning API to obtain a position.
- Motion Processing: Estimates the displacement from the IMU information
- Data Collection: Gathers and synchronise all the data
- Indoor Map request: Requests the maps of the nearby buildings for display and to improve the positioning.
- **Fusion (new)**: Improves the estimation of the current user's position and floor given all the information provided.
- **Map matching (new)**: Use the map to correct the position, avoid trajectories going through walls, and changing floors in the correct position.
- Test point marking: Allows the user to leave test point markers to verify the accuracy of the estimation
- Data Submission: Uploads the trajectory data to the openpositioning API
- **Data display (to be updated):** Displays the best estimate of the position of the user and its previous trajectory, position observation, measurements, etc.

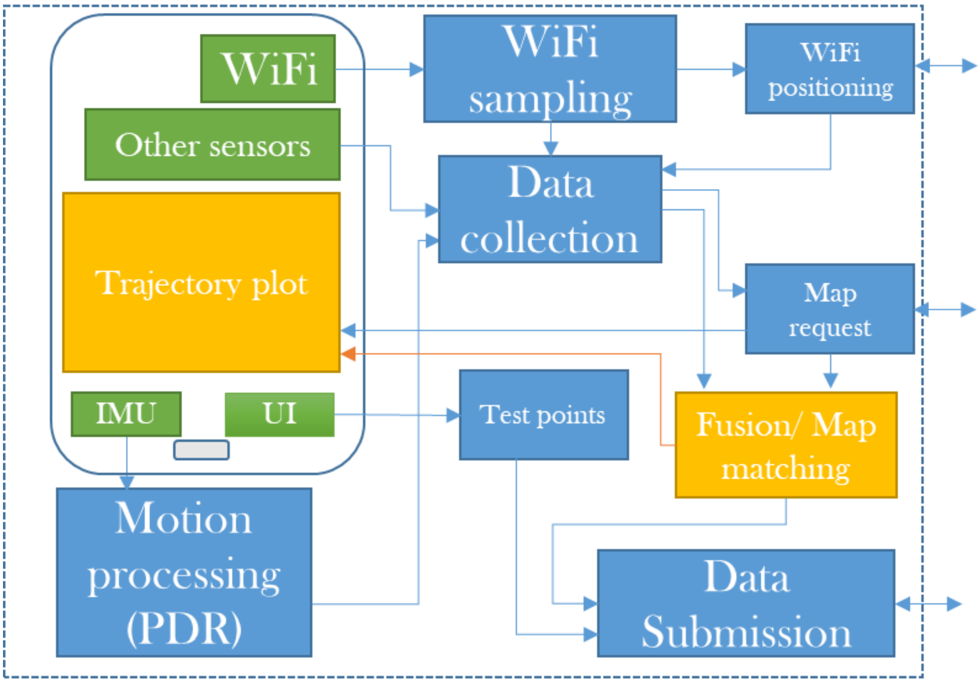

Figure 1. Basic diagram of the positioning app

Although most of the components will be Android programming, in some of the tasks we encourage using external tools – such as MATLAB or Jupyter Notebook - for visualization and prototyping. Also, some of the algorithms might be easier to test and debug in desktop environment (e.g. MATLAB, Java, or Python) before implementing them in the app itself.

We suggest that the students will work in the same groups as the 1st assignment. As the components have multiple subcomponents, we suggest that the teams decide and document different roles and responsibilities for different team members based on their interests and strengths, with some natural and necessary overlap. Some possible responsibilities include:

- UI design and implementation
- WiFi positioning
- Fusion model
- Map matching
- PDR motion model update
- WiFi positioning update
- Map information update
- Algorithm design and research
- Integration, evaluation, reporting

The course grading will be done by the course staff, and the top teams will be shortlisted to be evaluated by Huawei.

- ✓ The top teams' final evaluation session will take place in a workshop during April where each team will present their solution and contribute to the discussion and feedback from Huawei research engineers and scientists.
- ✓ The top performers will also receive certificates from Huawei in addition to top grades.
- ✓ The top individual contributors to the PositionMe app repository will also receive a certificate.

2.1 Indoor Environment

The test environment for the assignment will be the indoor routes and surroundings of the following buildings:

Figure 2. Areas of interest

- The Nucleus Building
- The Noreen and Kenneth Murray Library

The target of our data collection will be the Nucleus and the Noreen and Kenneth Murray Library (orange polygon), covering all their floors, entrances, main corridors, stairs and lifts (don't use the lift between the Nucleus and the Library in the ground floor), and entering the public access rooms to be able to map the area.

The openpositioning API will provide a WiFi localisation system for those buildings that will cover most of the main pathways. For an example of the openpositioning API refer to https://openpositioning.org/docs coarse and fine. For test scripts for the positioning refer to https://github.com/openpositioning/API_test_scripts

3 Core components

The app will have 2 new components and an update to the display, each representing different functionalities building on top of the current app. These are presented in order according to the dependency, so doing them in order is recommended. All the components need to be implemented in the phone, but you can extract data and show it using a desktop (in python, java, etc.).

3.1 Positioning fusion

- Implementing at least one sensor fusion algorithm
    - EKF
    - Particle filter (recommended for map matching)
- Initialise estimation (from GNSS, WiFi, others, no user selection)
- Use PDR movement model (work in the easting/northing space centred around a WPG84 point)
- GNSS positioning updates (WGS84, transform to the easting/northing space)
- WiFi positioning updates (from openpositioning, WGS84, transform to the easting/northing space)

3.2 Map matching

- Use the map of the building to improve the estimation of the position and floor of the user
- The map in the building has 3 types (indoor_type) of features per floor:
    - “wall”: represents areas of the building that cannot be crossed in that floor
    - “stairs”: steps to cross between floors, change height
    - “lift”: change between floors without significant horizontal displacement
- Use the movement model and the displacement to correct estimations that passes through walls
- Use the height change detected by the barometer to allow floor changes only near elevators/lifts
- Use the movement model and height change to distinguish between lift and stairs

3.3 Data display

- Display the best estimated position of the user near real time
- Smooth data filters on display options
- Display colour coded absolute position updates (last N observations)
    - GNSS
    - WiFi
    - PDR
- Display the full trajectory of fused positions updated every 1 second or when movement is detected

4 Grading

Each component will be graded based on the weighted criteria below.

4.1 Solution quality: 40%

The solution is evaluated for its functionality and results, but also for innovativeness. Don't be afraid to try out new approaches not covered in literature, and cite clearly libraries/solutions you ended up using or used as an inspiration. We will emphasize

- **Execution:** The assignment is mostly solvable with well-documented algorithms implemented in many libraries, but even here the integration and small details matter and learning to use such algorithms can be challenging.
- **Innovation:** Think outside the box, and try solve things your own way. Open your thought process and compare your solution to the ones presented in the literature.

To evaluate the accuracy of your solution you should record a video of a single live positioning session covering at least 3 floors of the Nucleus building and 1 floor of the library. We will record a trajectory with your app and set test points in known positions to estimate the algorithm error under real circumstances.

The required objectives will account for 80 % of the solution quality grading, and the accuracy will determine the additional 20 %.

4.2 Presentation: 20%

Presenting your results to show the process is often as important as the results itself. We are particularly interested in your problem solving, thought processes, and the solutions you come up with to different problems. Also, analysis on failed solutions and other hurdles is considered important. In your presentation, consider documenting at least the following:

- Problem statement and high-level solution description
- Application design choices and high-level code structure
- UI visualization
- Implementation issues and found solutions
- Algorithm selection and related literature
- Analysis of the problem and finding a solution (cite relevant literature)
- Solution success/failure analysis
- Performance analysis
- Data visualization

4.3 User Interface: 20%

Concentrate on functionality and intuitiveness. Good UI does not require a manual, but everything should be self-explanatory and work as expected. Apply and tune styles as the last touch. Document the UI decisions and implementation in your presentation/report.

4.4 Code quality: 20%

The code should reflect the functionality demonstrated in the presentation. The team should select a style described in a style guide and stick to it consistently [Style1, Style2]. Use of linters and other static code analysis tools (for example, PMD, StyleCheck, FindBugs [Code1]) is

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recommended to ensure consistent code style and quality. In the evaluation we will put strong emphasis on the following aspects:

- Consistency
- Descriptive variable/method/class naming
- Logically structured, modular code base
- Easy-to-follow, straightforward code flow
- Well-documented classes and methods
- Judicious, clarifying comments

Document the high-level structure in your presentation/report.

4.5 Code Submission

All work must be implemented in your own repositories forked from the base repository (https://github.com/openpositioning/PositionMe), of from your previous branches

- Provide your usernames and the forked repository link to your TA. The repository will need to be public at the end of the assignment
- Work on 1 or more (preferred) feature branches (Gitflow workflow) to divide the work on independent parts
- Make a pull request (use the "across forks" link) to merge your changes to the develop branch of the PositionMe repository, this will leave a record of your code version for review. The pull request must be made after 23:59 1st April 2026 and before 23:59 2nd April 2026, and no commits will be considered after 23:59 1st April 2026.

5 Reading and References

[Visu1] "How to Use Jupyter Notebook in 2020: A Beginner's Tutorial"

[PDR1] IPIN PDR Tutorial

[Data1] Multi sensor data collection

[Geom1] Mobility models for indoor spaces

[Code1] Static Code Analysis in Android

[Style1] Google Java Style Guide

[Style2] AOSP Java Code Style for Contributors